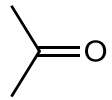
2(a)	**M1** Al_2O_3 is ionic **M2** P_4O_{10} is a (simple) molecule **M3** ionic bonds are stronger than intermolecular forces	**3**
2(c)(i)	increases rate by providing a different mechanism with a lower activation energy	**1**
2(c)(ii)	$2CH_3OH \rightarrow CH_3OCH_3 + H_2O$	**1**

Examiner report missing
9701_w25_qp_24 Question 2

3(a)(i)		**1**

3(a)(iii)	elimination	**1**

3(b)(i)	$(CH_3)_2CHOH + Na \rightarrow (CH_3)_2CHO^{(-)}Na^{(+)} + \frac{1}{2} H_2$	**1**

Examiner report missing
9701_w25_qp_24 Question 3

4(a)(i)	reduction	**1**

4(a)(ii)	PCl_5 / PCl_3 / $SOCl_2$ **OR** KCl / NaCl **AND** conc. H_2SO_4 / conc.H_3PO_4	**1**

Examiner report missing
9701_w25_qp_23 Question 4

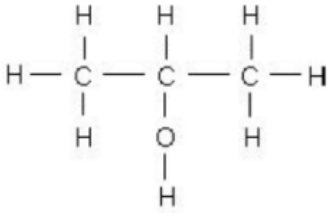
4(a)(i)	free-radical substitution	**1**
4(b)	sodium hydroxide **AND** water	**1**
4(c)(i)	orange to green	**1**
4(c)(ii)	$CH_3CH_2CH_2OH + 2[O] \rightarrow CH_3CH_2COOH + H_2O$	**1**

Examiner report missing
9701_w25_qp_22 Question 4

4(a)(i)	reduction	**1**

4(a)(ii)	PCl_5 / PCl_3 / $SOCl_2$ **OR** KCl / NaCl **AND** conc. H_2SO_4 / conc.H_3PO_4	**1**

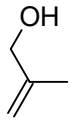
Examiner report missing
9701_w25_qp_21 Question 4

4(a)	$C_5H_{10}O_2$	**1**
4(d)(i)	$CH_3CN + HCl + 2H_2O \rightarrow CH_3COOH + NH_4Cl$	**1**
4(d)(ii)		**2**

Examiner report missing
9701_s25_qp_24 Question 4

5(a)(i)	C_2H_3O	**1**

5(a)(ii)	reagent	observation when reagent is added to W	structural formula of organic product when reagent is added to W	**4**
	Na	fizz / bubbles etc / **OR** solid / Na disappears / dissolves	$CH_2C(CH_3)CO_2Na$ **OR** $CH_2C(CH_3)CO_2^-Na^+$	
	Br_2 in the dark	brown to colourless / decolourise	$CH_2BrCBr(CH_3)CO_2H$	

5(b)(ii)		**1**

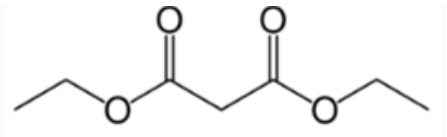
Examiner report missing
9701_s25_qp_23 Question 5

4(a)(i)	$LiAlH_4$	**1**
4(b)(i)	substitution	**1**
4(c)(i)	sp^3	**1**
4(c)(ii)	hydrogen / H_2	**1**
4(c)(iii)	(It / Na) reduces **A** / **B** / **C** / the alcohol (group) **OR** (it / Na behaves as) reducing agent **OR** (it / Na) donates electrons to **A** / **B** / **C** / the alcohol (group)	**1**
4(d)(i)	$(CH_3)_3COH$	**1**
4(d)(ii)	butan-1-ol **OR** (2) methylpropan-1-ol	**1**
4(d)(iii)	butan-2-ol **OR** $CH_3CH_2CH(OH)CH_3$.	**1**

Examiner report missing
9701_s25_qp_22 Question 4

4(a)	$C_3H_4O_3$	**1**
4(b)(i)	$(28.40 / 1000 \times 0.0005) = 1.42 \times 10^{-5}$ mol	**1**
4(b)(iv)	**M1** (also) oxidises / reacts with (primary and / or secondary) alcohol groups **M2** (also) breaks the sigma and pi bond in C=C	**2**

Examiner report missing
9701_s25_qp_21 Question 4

5(a)	**M1** two ester links **M2** rest of structure correct 	**2**
5(c)(i)	$HOOCCH_2COOH + Na_2CO_3 \rightarrow NaOOCCH_2COONa + CO_2 + H_2O$	**1**
5(c)(ii)	**C** is not as strong an acid as **B**	**1**
5(d)(i)	**M1** NaOH(aq) **AND** heat **M2** acidified **AND** potassium dichromate(VI) **AND** heat under reflux	**2**
5(d)(ii)	oxidation	**1**
5(e)(i)	$LiAlH_4$	**1**
5(e)(ii)	**M1** correct unambiguous organic product **M2** correct balanced equation $HOOCCH_2COOH + 8[H] \rightarrow HOCH_2CH_2CH_2OH + 2H_2O$	**2**
5(e)(iii)	1,3-dichloropropane	**1**

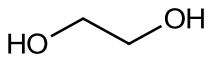
Examiner report missing
9701_s25_qp_21 Question 5

6(a)	carbonyl	**1**

6(c)(i)	**F** = propanone **G** = ethanal **H** = propanal	**2**

6(c)(iii)	$CH_3CH(OH)$-R where R is represented as H or any alkyl group	**1**

Examiner report missing
9701_s25_qp_21 Question 6

1(a)(i)	• (+)3 / III • –1 / I • (+)5 / V • –1 / I	**2**
1(b)(i)	green colour (of gas) disappears	**1**
1(b)(iv)	phosphoric(V) acid	**1**
1(b)(v)		**1**

Examiner report missing
9701_m25_qp_22 Question 1

4(a)			**5**

acidified $K_2Cr_2O_7$(aq)	(solution) turns (from orange to) green	**J** and **L**
alkaline I_2(aq)	yellow precipitate forms	**K** and **L**
2,4-dinitrophenylhydrazine / 2,4-DNPH	orange precipitate forms	**J** and **K**
Na(s)	effervescence	**L** and **M**

Examiner report missing
9701_m25_qp_22 Question 4

2(a)(i)	M1 $4Na + O_2 \rightarrow 2Na_2O$	**1**
	M2 $S + O_2 \rightarrow SO_2$	**1**

2(b)(i)	<table><tr><th>Period 3 oxide</th><th>product of rxn with water</th><th>pH of solution formed</th></tr><tr><td>● MgO</td><td>$Mg(OH)_2$</td><td>● $8 \leqslant pH \leqslant 12$</td></tr><tr><td>$P_4O_{10}$</td><td>● H_3PO_4</td><td>● $1 \leqslant pH \leqslant 4$</td></tr></table> ●✓●✓	**2**

2(b)(iii)		**2**

2(b)(iv)	alkyl groups are electron donating / have positive inductive effect.	**1**
	strengthens the O—H bond / makes H^+ less likely to be donated	**1**

Question 2

(a) (i) The formula of sodium oxide was not well known. In equations where correct formulae were shown, a significant proportion were not balanced.

(ii) Candidate answers were rarely given in terms of an aluminium cation and an oxide anion. When ions were described, they were generally shown with the appropriate number of outer electrons; charges on the ions were sometimes omitted or incorrect.

(iii) Weak candidate answers did not fully address the question and explained why the maximum oxidation state varies because of the variable number of valence electrons.

(b) (i) Many candidate answers referred to $Mg(OH)_2$ with a pH of a slightly acidic solution rather than weakly alkaline. The product of the reaction of P_4O_{10} and H_2O was not well known, $P(OH)_5$ was a common incorrect answer.

(ii) This equation proved to be demanding. Common errors included equations which were incorrectly balanced or showed ammonia as a product of the reaction.

(iii) This question was answered well by candidates. Occasionally the same structure was shown twice, usually with propan-1-ol as both answers.

(iv) Excellent candidate answers were seen and these explained why there were fewer $H^+(aq)$ in solution in terms of the increase in strength of O-H bond due to the positive inductive effect of the ethyl group. Inappropriate use of the term 'electronegativity' instead of 'electron density' was seen. Common misconceptions included the production of OH^- ions when alcohols dissociate or that water fully dissociates or that alcohols contain OH^- ions.

(c) (i) Many candidate answers referred to H_2O although this was not relevant in **(c)(i)**. Weaker responses appreciated that there was an increase in strength of intermolecular forces but did not explain how this change arose. Many candidates gave answers in terms of an increase in size of the molecules or in terms of nuclear attraction, nuclear charge and shielding, which is incorrect. A common misconception described the increase in boiling point due to an increase in strength of permanent dipoles as the group is descended.

(ii) Identification of hydrogen bonds in H_2O was relatively common. The relevant strength of hydrogen bonds in water compared to those intermolecular forces present in the other hydrides was not always included. Incorrect reference to H_2S containing hydrogen bonds was seen and there was ambiguity in answers that referred to the relative strength of the O-H bonds rather than intermolecular hydrogen bonds.

5(a)	cold dilute $KMnO_4$ **OR** cold dilute potassium manganate(VII)	**1**
5(d)(i)	**W** =	**1**
5(d)(iii)	$C_8H_{12}O_4 + 8[H] \rightarrow C_8H_{16}O_2 + 2H_2O$	**1**
5(d)(iv)	$LiAlH_4$	**1**

Question 5

This question tested organic reactions and spectroscopic methods using cyclohexene as a starting material. Many candidates seemed familiar with the chemistry, although the context of a cyclic molecule proved to be disconcerting for a few, who appeared to misinterpret the information given, or did not concentrate solely on the relevant functional groups.

(a) This was well answered by many, although differences appeared in the conditions to use with the $KMnO_4$ reagent (e.g. concentrated/dilute, alkaline/neutral/acidic).

(b)(i) Step 1 was familiar to many candidates. The hydrolysis of two functional groups in step 2 led many to think that H_2O_2 was being used, rather than two portions of NaOH.

(ii) In asking for a mechanism, 'electrophilic' was expected as well as 'addition'.

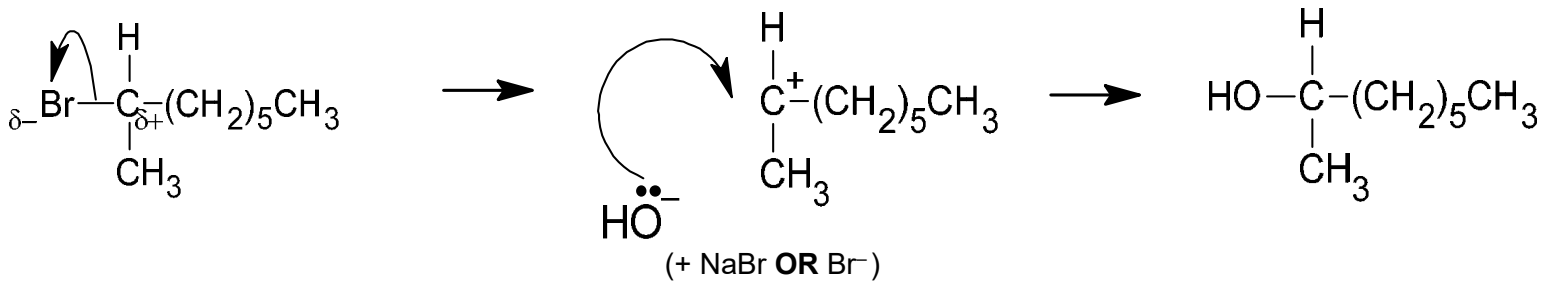
(c) Candidates showed some understanding in answering this question but often missed full credit by not following the rubric properly: the diagram needed labelling and each responsible bond (stretch) needed to be clearly identified. Many candidates only achieved half of this, or switched **F** and **G**.

(d)(i) This was mostly well answered, although some candidates lost credit when opting to give fully displayed versions of the CN group, neglecting to show a triple C≡N bond or using the wrong connectivity (N attached to the ring).

(ii) This was answered well by many candidates. Others did not acknowledge that acidic conditions are used, creating an ammonium species rather than NH_3.

(iii) The formulae of the organic products were generally well indicated, though candidates often then misbalanced the equation with insufficient (H).

(iv) This was generally well answered by candidates. A few $NaBH_4$ responses were offered, to no credit: carboxylic acids need the stronger of the two hydride reagents on the syllabus.

4(a)	Three-dimensional stereoisomers of $CH_3(CH_2)_5CHBrCH_3$ **M1** correct representation of left-hand answer as 3d structure of one of the enantiomers of $CH_3(CH_2)_5CHBrCH_3$ **M2** correct representation of 3d structure of the second enantiomer of $CH_3(CH_2)_5CHBrCH_3$	**2**
4(b)	 (+ NaBr **OR** Br^-) **M1** correct dipole ($^{\delta+}C—Br^{\delta-}$) **AND** curly arrow from bond of C—Br to Br (or just beyond) **M2** correct intermediate **ALLOW** R for $–(CH_2)_5CH_3$ **M3** curly arrow from lone pair on OH^- to the C of intermediate (or suitable position where C–OH bond is made) *If $SN2$ mechanism shown:* Apply $SN2$ mechanism mark scheme **only** if mechanism shows OH / NaOH species attacking the reactant and no OH **M1** correct dipole ($^{\delta+}C—Br^{\delta-}$) AND curly arrow from bond of C—Br to Br of reactant (or just beyond) **NOT M2** **M3** curly arrow from lone pair on $:OH^-$ to the appropriate C of the reactant	**3**

4(c)		**3**

Chemical test	observation with 2-bromooctane	observation with octan-2-ol	observation with $CH_3(CH_2)_5CHCH_2$
Br_2 in the dark	×		orange / brown to colourless
PCl_5		steamy fumes	×
$AgNO_3(aq)$	cream precipitate **OR** off white precipitate	×	

6 correct answers = 3 marks
4 or 5 correct answers = 2 marks
2 or 3 correct answers = 1 mark
0 or 1 correct answer = 0 marks

Question 4

(a) Some excellent representations of the three-dimensional structures were seen; these included the approximate bond angles for these tetrahedral structures. Other diagrams showed confusion over the types of symbols used to represent bonds in the plane with those in and out of the plane. In order to remove ambiguity, candidates should be encouraged to use the convention described in the syllabus, under expected conventions for representing organic structures.

(b) Good knowledge of the S_N1 mechanism was described with precision in the best answers. Some confusion regarding the relative charges on the C–Br dipole and either no reference to a lone pair of electrons on O of OH^- or the location of the lone pair of electrons on the H of OH^- were common incorrect details. Some answers confused the S_N2 mechanism details with S_N1.

(c) The observations of these reactions were generally well known. Some answers did not include observations, as directed in the table heading, rather than attempted to predict the product of the reactions.

(d) (i) The name of this reaction was quite well known. Addition, substitution, dehydration and cracking were common incorrect answers.

(ii) The reagent NaOH was described in many answers. Confusion of the relevant conditions was seen in weaker responses.

(e) (i) Fully correct answers were seen in around half of responses. Sometimes confusion between two of the types of hybridisations was seen. The weakest answers demonstrated a lack of understanding of the different types of hybridisations which occur in carbon atoms in order to produce single, double and triple covalent bonds.

(ii) It was common for answers to contain features which lead to both types of stereoisomerism rather than focus on the stereoisomerism present in unbranched hydrocarbons. A relatively small proportion of answers showed good understanding of the features of an alkene required in order for it to exist as geometrical (cis/trans) isomers. A lack of precision in answers produced some ambiguity in responses, for example ‘no movement of a double bond’ rather than ‘no rotation of a double bond’.

Question 5

(a) (i) Those candidates with a good understanding of mass spectrometry realised that the peaks shown in Table 5.1 represented the molecular ion peak, M^+, and its $(M+1)^+$ peak and used the appropriate formula to calculate the relative abundance of the M+1 peak. Some weaker responses based their calculation on an incorrect formula.

(ii) The strongest responses engaged with the information about compound **W**. Weaker responses did not engage fully with the question details and answered in terms of structural formulae. Some formulae gave the fragments with no charges or negative charges.

(b) (i) An understanding that an O–H bond, due to a carboxyl functional group, produces a broad stretch within the range 2500–3000 cm^{-1} was seen in the best responses. The weakest responses did not engage with the details given for compound **W** and its reaction to produce **X**, and listed every potential functional group which might account for bond **A** and bond **B**, including amide and ester functional groups.

(ii) Of those candidates that correctly identified **B** as a C=O of a carbonyl group, some candidates ignored the reaction conditions stated in the question and represented **X** as an aldehyde rather than a ketone.

(c) (i) A significant proportion of candidates correctly named the functional group present in **Y**, based on the information provided.

(ii) Many equations were shown with the correct formula of the organic product; not all equations were balanced. Identification of H_2O rather than H_2 as the other product was seen.

(iii) Those candidates who applied knowledge of the behaviour of different types of alcohol and carbonyl compounds correctly deduced the structure of **Y**. Confusion about the chemical reactions of alcohols and carbonyl compounds was seen in weaker responses.

6(a)(i)	(molecules with the) same structural formula (and same molecular formula) with different arrangement of atoms / groups in space	**1**

6(c)(i)	**M1** deduce n = 12 (from **Y** – 1C (in methanol)) **M2** $(100 \times x) \div (1.1 \times 100) = 12$ (so) $x = 13.2$	**2**

6(c)(ii)	<table><tr><th></th><th>observation on addition of 2,4–DNPH</th></tr><tr><td>**Y**</td><td>orange precipitate</td></tr><tr><td>**Q**</td><td>no precipitate</td></tr></table> Both correct for one mark	**1**

6(c)(iii)	$ROH + Na \rightarrow RONa + 1/2H_2$ (0.002 mol **Q** produced 0.002 mol H_2 gas so) 2 OH groups **M1** *answer indicates that OH group(s) in **Q** react with Na to produce the H_2 in the ratio 1 mol OH : ½ mol H_2* **M2** *uses data to show 2OH groups*	**2**

Question 6

(a) (i) The definition of stereoisomerism was one of the least well answered questions. The most common error was to state that 'molecules had the same molecular formula but different structural formula'. Few mentioned that it was the arrangement of the atoms or groups in space.

(ii) Very few answers gained full credit here. Some candidates identified a C=C double bond and two chiral centres and then incorrectly stated that this gave (2 × 3 =) 6 stereoisomers instead of using the correct expression 2^3 (= 8).

(iii) The molecular formula of **V** was answered quite well. There were no common errors.

(iv) Many candidates correctly identified an alkene and carbonyl group. The most common error was to omit the ester group and substitute it for an alcohol group or a carboxyl/carboxylic acid group.

(b) (i) The question required an identification of the role of reagent **T**. Candidates who stated that **T** behaved as a reducing agent gained full credit. Those candidates who ventured to explain the behaviour of **T** with each functional group generally understood that it reduced the carbonyl group in **V** to an alcohol group in **W**. The reaction of the C=C to from a saturated chain attracted a variety of incorrect answers ranging from acting as a catalyst to hydrolysing the alkene group.

(ii) This question was well answered.

(c) (i) Candidates found this very challenging. Many answers used the m/e values of 200 and 201, from the mass spectrum, instead of the relative abundance values of 100 and x, and attempted to average these values to determine the relative atomic mass.

Creditworthy answers deduced that **Q** contained 12 C atoms and then used this value of *n* in the expression $n = \dfrac{100 \times \text{abundance of M} + 1 \text{ ion}}{1.1 \times \text{abundance of M} + \text{ion}}$

to calculate the abundance of the (M + 1) ion which is equivalent to the value of x.

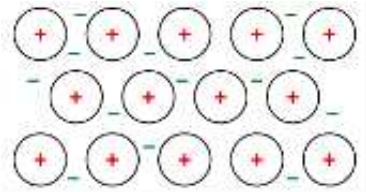
(ii) This question was answered correctly by many candidates with the most common error being to omit 'precipitate' with the orange colour for the observation with 2,4-DNPH with **Y**.

(iii) The majority of candidates did not recognise that the volume of hydrogen gas was collected at s.t.p., in the question, which, from the Table of data, is 22.4 $dm^3 mol^{-1}$.

Various methods were used to calculate the moles of gas produced to include the ideal gas equation and 24.0 $dm^3 mol^{-1}$.

Those candidates who managed to calculate the number of moles of gas then needed to appreciate that the ratio of hydroxyl groups to hydrogen gas was 2 : 1. The award of full credit was rare.

(iv) Many candidates gave correct absorption ranges for one of the two carbonyl absorptions in **Y**, for the ketone and ester functional groups. Several answers gave a range of absorptions for a carboxyl group in **Q** as a difference in the infrared spectrum, having not appreciated that **Q** contains **only** hydroxyl groups, mentioned in the stem of **(c)(ii)**.

1(a)	 labelled diagram shows: • regular arrangement of positive ions • **AND** surrounded by / (sea of) delocalised electrons	**1**

1(d)(i)	a substance that removes electrons from another / causes electron loss / gains electrons	**1**

1(d)(ii)	($KMnO_4$ is) **cold** and **dilute**	**1**

Question 1

This was a broad-based question, encompassing ideas of physical, inorganic and organic chemistry. Candidates were adept with simple definitions, but application of ideas showed some deficit of understanding.

(a) Many diagrams were well presented, but a substantial number neglected to label the particles: mention of delocalisation of electrons was crucial. A sizable minority incorrectly identified the positive particles as protons or nuclei.

(b) (i) This question was mostly well answered, though errors of balancing and 'Bi_2' were often seen.

(ii) Although many candidates secured full credit here, common errors were not to link deduction and reasoning (e.g. giant *because* high melting point) and to offer contradictory statements (e.g. giant molecular and ionic bonding). In many cases, brevity served better. Some candidates appeared confused about the compound they were being asked to describe, Bi_2O_3, rather than bismuth itself.

(c) (i) This was answered well, though answers of +6 were commonly seen with respect to bismuth.

(ii) Many different answers were offered here, including $NaBH_4$ and O_2, whereas candidates needed to refer to a compound given in the equation.

(d) (i) This was generally well answered. It is preferable to refer to transfer of electrons when explaining the term oxidising agent.

(ii) Some contradictions were observed here, such as 'cold dilute … heat under reflux.'

(iii) Many candidates answered this well. Common errors included a switch of answers in the case of Tollens' reagent and neglecting to mention a precipitate or solid in the iodoform and DNPH reactions.

(iv) Many candidates tried unsuccessfully to include $NaBH_4$ in their answers. Candidates are reminded that even with the use of [H], there is a need for a stoichiometric balancing number. Use of CH_3COH is discouraged as a method of representing aldehydes.

(e) The initial stages of most calculations were good, but many candidates then forgot to multiply by 40 (to scale up to 1 dm^3 from 25 cm^3) or to multiply the concentration by 10 (1000 ÷ 100). In general, working needed to be shown more clearly, as some inventive methods could otherwise have given a seemingly correct answer.

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(e)(i)	1 4 0	**1**

3(e)(ii)	Solution turns from purple to colourless	**1**
	Orange / red precipitate forms	**1**
	no reaction	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

3(a)(i)	$4P / P_4 + 5O_2 \rightarrow P_4O_{10}$	**1**
	$P_4O_{10} + 6H_2O \rightarrow 4H_3PO_4$	**1**

3(c)(i)	H_2O / steam	**1**

3(c)(ii)		**1**

Question 3

(a) (i) Fully balanced equations were seen by more able candidates.

(ii) This definition was described accurately in most candidates answers. A common misconception was seen in weaker responses that described a weak acid as one that *'did not completely dissolve'*.

(b) (i) Many candidates gave a definition of the enthalpy change which occurs when 'one mole of product is formed from its elements'. It was less common for the elements to be described as 'in their standard states'.

(ii) More able candidates gave excellent and logical workings that contained clearly labelled energy cycles. It was common to see an expression for the different bond energies added or subtracted from each other, in a variety of possible incorrect combinations, with no attempt to show how these expressions were deduced. The stoichiometry of the equation was ignored by some.

(iii) Excellent answers were seen which identified phosphorus in H_3PO_3 as both oxidised and reduced and linked the process of oxidation and reduction to the relevant changes in oxidation number for the three phosphorus containing species. Weaker responses were less specific, for example '*H_3PO_3 is oxidised and reduced'*. Oxidation numbers were sometimes described incorrectly.

(c) (i) Some candidates identified **A** as steam or $H_2O(g)$, other candidates were less precise and identified **A** as water. A common incorrect answer was NaOH.

(ii) Many unambiguous representations of 2-bromopropane were seen. Occasionally the central carbon atom was drawn with a valency of 3.

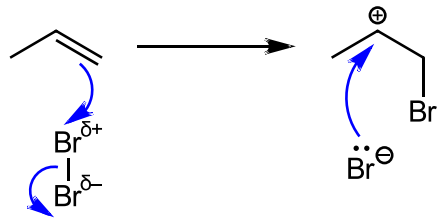
(iii) Description of this esterification reaction as a condensation reaction was relatively well known. Incorrect answers included hydrolysis and oxidation.

(iv) This question proved to be demanding. Candidates were expected to ignore all bonds which were common to both reactant and product. While the appropriate bond, C=O, was identified in many answers C-O and C-H were also seen. Most candidates only identified a range of wavenumbers where a change in absorption occurs rather than described if the absorption appeared, disappeared or moved.

(d) (i) More able candidates deduced a formula for the correct organic product in the equation. The inclusion of H_2O as the additional product was less common, even when included the equation was not always balanced.

(ii) The appropriate formula to deduce the number of carbon atoms present in **T** was not well known and many confused calculations were seen. It was common for candidates to give a structural formula for **T** rather than the required molecular formula.

4(a)(i)	(it has molecules that) contain a chiral carbon / centre **OR** are non-superimposable mirror images	**1**

4(b)(i)	 curly arrow from C=C bond to $Br^{\delta+}$ **AND** curly arrow from $Br^{\delta+}$— $Br^{\delta-}$ bond to $Br^{\delta-}$ **AND** $Br^{\delta+}$ closest to the double bond	**1**
	correct intermediate	**1**
	curly arrow from lone pair on Br^- to $C^{(+)}$ of the intermediate	**1**

4(b)(ii)	$CH_3CH(OH)COOH + [O] \rightarrow CH_3COCOOH + H_2O$	**1**

Question 4

(a) (i) Some candidates gave clear explanations of the presence of a chiral carbon in the lactic acid structure. A few candidates stated that optical isomers had chiral centres or described the behaviour of an optical isomer in the presence of polarised light.

(ii) Many candidates were unable to give the systematic name of lactic acid. 'Carboxylic acid' was seen frequently.

(iii) It was common for diagrams to show **more** than one water molecule or **multiple** hydrogen bonds. Candidates should be advised to respond as directed by the question. It was quite common for answers, showing more than one hydrogen bond, to contradict a correct representation.

(b) (i) Some clear and precise reaction mechanisms were shown with partial charges assigned to relevant atoms and unambiguous representation of curly arrows to show bonds broken and bonds made. When drawing mechanisms of this type candidates should be encouraged to consider what the curly arrows represent and whether their representation is clear, use partial charges correctly and discriminately and check that the number of carbon atoms on the intermediate is correct.

(ii) The equations described sometimes contained only pyruvic acid as a product or identified H_2 rather than H_2O as a product.

(iii) This question was demanding. Many candidates confused the reagents and conditions required to convert halogenoalkanes to nitriles with those required to convert ethanal to a hydroxynitrile. Deduction of the reagents and conditions used in the second step of both synthesis from propene and ethanal proved demanding for most candidates. Oxidising agents were more commonly seen for the third steps, however, some candidates represented the oxidising agents with incorrect formulae or with no reference to acid conditions.

3(a)	compound / molecule / substance that oxidises another / is reduced	**1**

3(e)(i)	1 4 0	**1**

3(e)(ii)	Solution turns from purple to colourless	**1**
	Orange / red precipitate forms	**1**
	no reaction	**1**

Question 3

(a) The definition of an oxidising agent, as a species that oxidises another species, and is reduced in the process by the gain of electrons, was overall generally well answered.

(b) This question required the completion of an energy diagram/energy profile to represent the catalytic effect of MnO_2(s) on the decomposition of $KClO_3$(s). This entailed drawing two labelled curves representing the uncatalysed and the catalysed profiles.

Many answers were awarded full credit. Common errors included unlabelled diagrams or profiles with only one curve.

(c) The term *disproportionation* is used to describe the oxidation and reduction of a single element, in the same chemical reaction. Here the reaction involved $KClO_3$ being converted to $KClO_4$ and KCl, in which the Cl is both oxidised and reduced in this reaction.

Many chose the correct element; there were, however, several answers that identified K or $KClO_3$ as the species undergoing simultaneous redox.

(d) (i) PV=nRT was successfully used by many candidates to calculate the volume of 1 mole of gas, in a reaction of glucose with molten $KClO_3$, to give the correct answer of 0.0524 m^3, to 3 significant figures.

$$4KClO_3 + C_6H_{12}O_6 \rightarrow 6CO_2 + 6H_2O + 4KCl$$

Several candidates did not appear to read the question carefully and gave answers with an incorrect number of sig. figs and/or incorrect rounding of their answers.

(ii) This question required the calculation of the moles of glucose, (5/180) and to use this value, together with their answer from 3**(d)(i)**, to calculate total volume of the gaseous CO_2 and H_2O in the equation.

Many candidates were able to correctly calculate the moles of glucose but then failed to multiply this value by 12 to determine the total moles of CO_2 and H_2O produced.

Several answers then continued to repeat the calculation in 3**(d)(i)** instead of using their answer from 3**(d)(i)**.

(e) (i) A high proportion of candidates correctly identified the number of primary, secondary and tertiary alcohol groups, in glucose, as 1, 4 and zero, respectively.

(ii) This question required observations for three tests.

The observation for a reaction of glucose + warm, acidified $KMnO_4$(aq) is that the solution changes from purple to colourless.

Many candidates were given credit for a correct observation.

Glucose + warm, Fehling's reagent gave an orange/red precipitate which again was recognised by many candidates. Several candidates omitted the precipitate or stated that a silver mirror formed.

Glucose + warm, alkaline I_2(aq) was incorrectly stated as a yellow precipitate by many candidates. There is no reaction with glucose, so the solution remains colourless.

(iii) Several candidates were able to correctly define *structural isomers* as 'molecules with the same molecular formula but different structural formulae'.

5(a)	C_3H_7O	**1**

5(d)				**3**

reagent	observation on addition to **X**	observation on addition to **Y**
sodium carbonate solution	effervescence	no change
2,4-DNPH	orange precipitate	no change
alkaline aqueous iodine	yellow precipitate	yellow precipitate

Two correct for one mark, four correct for two marks, six correct for three marks

Question 5

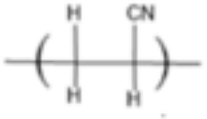
(a) Some candidates deduced the correct empirical formula of C_3H_7O from a molecular formula of $C_6H_{14}O_2$.

(b) (i) Very few candidates used molecular formula, as required in the question, to construct an equation for the reduction reaction.

(ii) Many answers correctly stated that a suitable reducing agent for the reaction is $LiAlH_4$.

(c) Some good answers were seen.

(d) Many candidates correctly stated that **X**, the ketoacid, would react with all three reagents to give effervescence, an orange precipitate and yellow precipitate respectively. Fewer candidates recognised that the diol, **Y**, would only react with the alkaline $I_2(aq)$ as it contained a $CH_3CH(OH)$ moiety.

6(a)	 OR 	**1**
6(b)(i)	nucleophilic addition	**1**
6(b)(iii)	heat + conc. sulfuric acid / conc. phosphoric acid **OR** heat + Al_2O_3	**1**
6(f)(i)	substitution	**1**

Question 6

(a) The single repeat unit derived from the monomer, CH_2=CHCN, by addition polymerisation was quite well answered.

(b) (i) Many candidates recognised that the mechanism for step 1 was nucleophilic addition.

(ii) Overall, the mechanism for step 1 was generally not answered well. The polarity on the C=O bond was often absent and the arrow from the CN^- to the C (in the C=O bond) did not start from the lone pair on the C of the CN^- ion. Several answers contained an incorrect intermediate with a positive charge which was then attacked by the CN^- nucleophile and many answers incorrectly showed an arrow from an H^+ ion to an O^- (in the intermediate).

(iii) Some good answers were seen.

(iv) The absorptions in the infrared spectrum of CH_2=CHCN were often correctly identified as C≡N and C=C. Incorrect answers mainly contained an identification of the functional groups rather than the bond or identifying the absorption (annotated as T on the IR spectrum) as that of a C=O group.

(c) (i) Very few candidates gave an accurate description.

(ii) A number of answers were awarded some credit. Few candidates were able to give a full description.

(d) Several candidates gave a correct pair of molecules that were non-superimposable images of $CH_3CH(OH)CN$. A common error was diagrams containing the same type of bond at 180° to each other.

(e) The equation for the reduction of propanenitrile to propylamine using hydrogen gas was not well known. Few candidates used the correct stoichiometry for hydrogen.

(f) (i) It was well known that the name of the type of reaction in step 1 is substitution.

(ii) Many candidates gained credit for ammonia. Many omitted heat or the solvent as conditions.

3(a)(i)	KCN in ethanol + heat	**1**

3(b)	acid (conditions) / H^+(aq) **AND** heat under reflux	**1**

Question 3

(a) (i) Some candidates incorrectly described the reagent and conditions required to convert carbonyl compounds to hydroxynitriles, rather than halogenoalkanes to nitriles.

(ii) The systematic name for **X** was not always used, e.g. propionitrile or ethyl cyanide. Omission of the 'e' in propanenitrile was often seen.

(iii) It was generally well known that this type of reaction is a hydrolysis reaction.

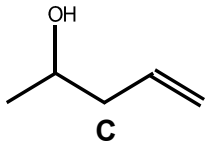
(iv) Many candidates correctly produced equations representing the production of hydrogen gas. The formula of the calcium salt was frequently given in terms of calcium with a valency of one.

(b) Many candidates knew that acidic conditions were required. Stronger candidates appreciated that the essential condition for this reaction was 'heat under reflux' to ensure complete oxidation to the carboxylic acid.

(c) Most candidates described the correct number of pi bonds present in **X**; the correct number of sigma bonds was seen less frequently. Seven, nine and ten sigma bonds were common incorrect answers.

5(a)	molecules / isomers with the same molecular formula / same number of atoms of each element **AND** different structural formulae / different structures	**1**

5(b)(i)	carbonyl (group)	**1**

5(c)(i)	 **C** **M1** $CH_3CH(OH)$ present **M2** five carbons in a straight chain **AND** molecular formula $C_5H_{10}O$ **AND** 1 C=C structure which shows only one type of stereoisomerism	**2**

Question 5

(a) This definition was generally well known.

(b) (i) Many candidates appreciated that the two molecules were members of the carbonyl functional group. Some candidates incorrectly referred to them as either aldehydes or ketones.

(ii) Correct structures of **A** were common. Many candidates represented **B** as a ketone; the correct carbon skeleton was not always deduced.

(c) (i) It was common for some of the information in the question to be used to produce partially correct structures. Many candidates incorrectly produced structures with two types of stereoisomerism.

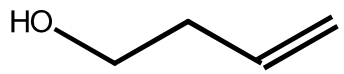
(ii) This was generally well answered.

(d) (i) Ketone and carboxylic acid were the most common incorrect answers.

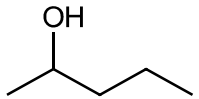
(ii) Hydrolysis was a common incorrect response.

(iii) Deduction of the structure of **D** proved to be demanding. Often the structure drawn had a carboxylic acid group at one end of the molecule; identification of $(CH_3)CH(OH)$– at one end of the molecule was seen less frequently. Occasionally, the correct ends of the molecule were drawn with an incorrect number of carbon atoms in the carbon backbone.

(iv) Stronger candidates clearly identified two differences between the spectra in terms of the different absorption ranges produced as a result of the different environment of the C=O bond in an ester compared to a carboxyl group, or in terms of the presence of the O-H bond in a hydroxyl or carboxyl group which is absent in the ester, **E**. Reference to absorbance at 2850 – 2950 due to C-H bonds was common, even though both **D** and **E** contain this type of bond.

4(a)(i)		**1**

4(b)(i)	cold dilute acidified $KMnO_4$	**1**

4(c)	 **M1** molecule with 5C atoms and 1 alcohol group **M2** molecule with —$CH(CH_3)OH$	**2**

Question 4

(a) (i) Common errors were to retain the COOH group and to have the C=C double bond reduced.

(ii) This was well answered by most candidates.

(b) (i) This was generally well answered, although few candidates stated all the required conditions.

(ii) This was well answered by most candidates.

(iii) A small proportion of candidates gained full credit here.

(c) Many candidates correctly gave pentan-1-ol. A small number of candidates made the link between the yellow precipitate with alkaline I_2(aq) and a $CH_3CH(OH)$– group. Many candidates struggled to synthesise all the detail to present a cogent answer to the question.

3(a)(i)	(a reaction where) two (or more) compounds / reagents / molecules form (only) **one product**	**1**
3(a)(ii)	position(al isomerism)	**1**
3(a)(iv)	96% × (5000 / 42.0) × 72.0 = 8230 or 8229 (kg)	**1**
3(b)(iii)	$C_4H_8O + [O] \rightarrow C_4H_8O_2$	**1**
3(c)	*alkane* C_3H_8	**1**
	alcohols $C_4H_{10}O$	**1**

Question 3

This question was largely framed around organic chemistry, with the inclusion of some physical and stoichiometric work. It was well attempted on the whole, although knowledge of some key definitions was seen to be weaker.

(a) (i) Many candidates appreciated that a single product was the outcome of an addition reaction, but many others described/attempted to describe an addition reaction without indicating that only one product was formed.

(ii) This was well answered by most candidates.

(iii) The major errors here were butan-1-one, butan-1-ol/butanol or propanal/propan-1-al. Candidates are reminded that there is no need to offer a locational number for the aldehyde group.

(iv) Very few candidates converted the quantities to moles, so simply offered 96% of 5000 kg as the answer.

(b) (i) Candidates seemed largely unaware of the nucleophilic addition mechanism. Common errors and omissions included:

- incorrect placement of the lone pair of electrons on the N of the CN^- ion.
- giving the intermediate a positive charge which was subsequently attacked by the CN^- nucleophile.
- mechanisms involving the H (of HCN) / H^+ attacking the intermediate negative ions.

- curly arrows inaccurately coming from the C (in the C=O group).
- the H (of HCN) shown attacking the O (in the C=O group) to form an intermediate hydroxyl derivative with a positive charge.

(ii) The first and third marks of this question gave the largest number of variants, with Fehling's reagent or DNPH offered for experiment 1. For experiment 3, many answers erroneously gave a change or loss of colour rather than a method of noting that no reaction had occurred.

(iii) There were some very good answers here, although some increased the number of H atoms in the product and/or H_2 appeared as a product.

(iv) This was generally very well answered, with occasionally some candidates circling two C atoms or the entirety of HOCHCN.

(c) It appeared that many candidates did not understand the term molecular formula and gave structural formulae throughout.

(d) (i) Some answers referred to a change in position of the equilibrium/yield, which was not credited.

(ii) Many answers excluded the word gas and just stated 'less/fewer (number of) moles'. Although the reaction in question was entirely in the gas phase, precision in an answer is crucial. A few candidates gave the incorrect direction of the equilibrium shift.

(iii) Many candidates used a value of 436 in their answers, which suggests the use of a Data Booklet. Centres are reminded that Data Booklets are **not** to be issued to candidates for these papers.

(iv) Any mention of hydrogen bonding was not credited, in accordance with the convention and definition used by this qualification.

4(a)(i)	NaOH in alcohol / ethanol AND heat (under reflux)	**1**

4(a)(iii)	$C_4H_{10}O$ / $CH_3CH(OH)C_2H_5$ → C_4H_8 / $CH_3CHCHCH_3$ + H_2O	**1**

4(b)(ii)	CHI_3 / iodoform / tr(i)iodomethane	**1**
	$CH_3CH_2CO_2^-$ / propanoate	**1**

Question 4

This was a second organic chemistry question, which included some analytical work. This was generally answered well by many candidates, though later parts on IR and MS appeared less familiar to some.

(a) (i) 'Heat' or equivalent was often missing. H_2SO_4 was a popular answer despite the conditions suggested in the syllabus.

(ii) Many repeat units were shown containing four carbon atoms, as if the CH_3 groups had been incorporated into the backbone. Some candidates showed an ethene structure with missing CH_3 side groups.

(iii) Candidates very often showed the reverse equation with C_4H_8 and H_2O. Quite a few answers contained Al_2O_3 as a reactant and several answers omitted H_2O as a product.

(b) (i) This was well answered.

(ii) There were many correct structures/names for iodoform but very few correct propanoate ions.

(c) (i) There were many correct answers to this item.

(ii) Catalyst and oxidising agent were the common errors here, but otherwise there were many correct answers.

(d) (i) Compound **F** was commonly given incorrectly as the unknown compound, referring to an O–H absorption in the range 2850–3000 cm^{-1}. Candidates are reminded that C–H stretches appear in all organic spectra. Where absorptions are referenced, candidates need to identify which bond is stretching: 'nitrile', for example, is insufficient.

(ii) There were many null responses to this. Where attempted, candidates' answers were evenly split between correct identification of an identical abundance and some complicated (erroneous) mathematics.